HYUNJE YANG

301 E Dean Keeton St, Austin, Texas, USA | +1 512 228 6214
hy7555@my.utexas.edu | https://hyunjeyang.com

Short version. Full CV available at hyunjeyang.com

EDUCATION

The University of Texas at Austin (UT Austin) — **Austin, TX, USA**
Ph.D. Candidate in Civil, Architectural and Environmental Engineering — *Aug. 2023 – Present*
- *UT Austin Engineering Fellowship, Cockrell School of Engineering and UT Austin Graduate School*
- *Teaching Assistant, Probability and Statistics for Civil Engineers (Spring 2025, 2026)*

Seoul National University (SNU) — **Seoul, South Korea**
M.S. in Forest Environmental Science — *Mar. 2016 – Feb. 2018*
- *Thesis: Formula for Calculating Mean Velocity in Mountain Streams Using the Salt Dilution Method*
- *Teaching Assistant, Forest Geological Information (Spring 2016)*

B.S. in Forest Environmental Science — *Mar. 2012 – Feb. 2016*
- *National Scholarship Award for 7 semesters, Ministry of Science and ICT*

RESEARCH INTERESTS

- Transferable AI Modeling
- Forecast Uncertainty and Early Warning
- Compound Coastal Flooding
- Hydrology of Ungauged Systems

RESEARCH EXPERIENCE

Oak Ridge National Laboratory (ORNL) — **Oak Ridge, TN, USA**
Graduate Research Intern — *May 2026 – Jul. 2026*
Research Collaborator · Austin, TX, USA (Remote) — *Jul. 2026 – Present*
- Geometry-aware neural operators to scale storm-surge prediction from local coastal domains toward regional-to-global coverage on unstructured meshes
- Investigating coastal wetland resistance and recovery using global tropical-cyclone records and satellite observations
- Developing a diffusion-based surrogate for reconstructing historical storm-surge dynamics

The University of Texas at Austin (UT Austin) — **Austin, TX, USA**
Ph.D. Student, Civil, Architectural and Environmental Engineering — *Aug. 2023 – Present*
- Mask-Aware Multimodal Deep Operator Network for Spatially Contiguous Mapping of Soil Water Retention Curves
- Texas-Scale Assessment of Multi-Driver Compoundedness in Coastal Flooding
- Physics-Informed Machine Learning for Compound Flooding Prediction
- Integration of Machine Learning and Static Flood Models for Coastal Inundation Simulation
- Reconstruction of Missing Data for Hurricane-Damaged Buildings using XGBoost

National Institute of Forest Science (NIFoS) — **Seoul, South Korea**
Researcher, Forest Environment and Conservation Department — *Mar. 2018 – Apr. 2023*
[Government R&D Projects]
- Establishment of Big Data and Integrated Utilization System for Flash Floods in Forest Watershed
- Development of Forest Water Management Technology to Reduce Stream Depletion and Nonpoint Source Pollution
- Quantification and Improvement of Forest Water Yield for Sustainable Supply of Freshwater
- Long-term Monitoring of Flow Characteristics in Different Forest Stands and Locations

Seoul National University (SNU) — **Seoul, South Korea**
Researcher, Department of Forest Environmental Science — *Jun. 2013 – Feb. 2018*
- Hydraulic Relation of Discharge and Velocity in Mountain Streams Using Salt Dilution Method
- Monitoring of Floods in Mountainous Areas and Critical Runoffs in Urban Areas

PUBLICATIONS

[Under Review]

- **Yang, H.**, Lee, J.W., Lee, W. "Where, When, and How Do Physics-Informed Deep Learning Models Improve Hurricane-Induced Compound Flooding Predictions?" *ESS Open Archive*, 2026. https://doi.org/10.22541/essoar.15005345/v1

[In Preparation]

- **Yang, H.**, Li, Z., Lee, J.W., Kim, T., Tran, V.N., Lu, D., Dias, P.A., Painter, S.L., Le, P.V.V. "Geometry-Aware Neural Operators for Transferable Storm Surge Prediction" *In preparation.*
- Li, Z., **Yang, H.**, Dias, P.A., Faxon, A., McCarthy, M., Sulman, B., Painter, S.L., Tran, V.N., Lu, D., Le, P.V.V. "Divergent Controls on Coastal Wetland Resistance and Recovery to Tropical Cyclones" *In preparation.*
- Le, P.V.V., Dias, P.A., **Yang, H.**, Li, Z., Tran, V.N., Lu, D., Faxon, A., McCarthy, M., Sulman, B., Painter, S.L. "Learning Transferable Storm Surge Dynamics from Synthetic Hurricanes Using Diffusion Models" *In preparation.*
- Chowdhury, M.E., **Yang, H.** "Analysis of Flood-Driver Transition across 14 Historical Storms in the Houston–Galveston Region" *In preparation.*
- **Yang, H.***, Kim, T.*, Lee, J.W., Ivanov, V. "HT-PTF: Landscape-Informed Operator Learning for Spatially Contiguous Mapping of Soil Water Retention Curves" *In preparation.* (*Both authors contributed equally to this work)

[Journals]

- **Yang, H.**, Lee, J.W., Santos Cruz, A.U. "Machine Learning-Enhanced Static Flood Models for High-Resolution Peak Storm Surge Inundation Mapping in Southeast Texas, USA" *Journal of Hydrology: Regional Studies*, 2026, 63: p. 103056. https://doi.org/10.1016/j.ejrh.2025.103056
- **Yang, H.**, Lee, J.W., Klepac, S., Santos Cruz, A.U., Subgranon, A., Jiao, J. "Reconstructing Missing Data of Damaged Buildings From Post-Hurricane Reconnaissance Data Using XGBoost" *Frontiers in Built Environment*, 2024, 10: p. 1444001. https://doi.org/10.3389/fbuil.2024.1444001
- **Yang, H.**, Lim, H., Moon, H., Li, Q., Nam, S., Choi, B., and Choi, H.T. "Identifying the Minimum Number of Flood Events for Reasonable Flood Peak Prediction of Ungauged Forested Catchments in South Korea" *Forests*, 2023, 14(6): p. 1131. https://doi.org/10.3390/f14061131
- Nam, S., **Yang, H.**, Lim, H., Kim, J., Li, Q., Moon, H., and Choi, H.T. "Short-Term Effects of Forest Fire on Water Quality Along a Headwater Stream in the Immediate Post-Fire Period" *Water*, 2023, 15(1): p. 131. https://doi.org/10.3390/w15010131
- **Yang, H.**, Lim, H., Moon, H., Li, Q., Nam, S., Kim, J., and Choi, H.T. "Simple Optimal Sampling Algorithm to Strengthen Digital Soil Mapping Using the Spatial Distribution of Machine Learning Predictive Uncertainty: A Case Study for Field Capacity Prediction" *Land*, 2022, 11(11): p. 2098. https://doi.org/10.3390/land11112098
- **Yang, H.**, Yoo, H., Lim, H., Kim, J., and Choi, H.T. "Impacts of Soil Properties, Topography, and Environmental Features on Soil Water Holding Capacities (SWHCs) and Their Interrelationship" *Land*, 2021, 10(12): p. 1290. https://doi.org/10.3390/land10121290
- Lim, H., **Yang, H.**, et al. "Development of Pedo-Transfer Functions for the Saturated Hydraulic Conductivity of Forest Soil in South Korea Considering Forest Stand and Site Characteristics" *Water*, 2020, 12(8): p. 2217. https://doi.org/10.3390/w12082217
- **Yang, H.**, Choi, H.T., and Lim, H. "Effects of Forest Thinning on the Long-Term Runoff Changes of Coniferous Forest Plantation" *Water*, 2019, 11(11): p. 2301. https://doi.org/10.3390/w11112301
- **Yang, H.**, Choi, H.T., and Lim, H. "Applicability Assessment of Estimation Methods for Baseflow Recession Constants in Small Forest Catchments" *Water*, 2018, 10(8): p. 1074. https://doi.org/10.3390/w10081074
- **Yang, H.**, Lee, S., and Im, S. "Hydraulic Relation of Discharge and Velocity in Small, Steep Mountain Streams Using Salt Dilution Method" *Journal of Korean Society of Forest Science*, 2018, 107(2): p. 158-165. https://doi.org/10.14578/jkfs.2018.107.2.158
- **Yang, H.**, Im, S., Lee, Q., Eu, S., and Lee, E. "Flow Measurement of Mountain Streams using Salt Dilution Method" *Journal of the Korean Society of Forest Engineering*, 2016, 14(1): p. 1-6. https://www.earticle.net/Article/A276485

[Books]

- Choi, H.T., Kim, J., Lim, H., and **Yang, H.** *Standard Manual for Developing Forest Soil Water Map.* National Institute of Forest Science: Seoul, South Korea, 2021. (ISBN: 9791160195859)
- Choi, H.T., Kim, J., Lim, H., **Yang, H.**, and Yoo, H. *Long-term Monitoring of Forest Catchment Runoff Characteristics by Different Forest Sites and Stands.* National Institute of Forest Science: Seoul, South Korea, 2021. (ISBN: 9791160195514)
- Choi, H.T., Kim, J., Seol, A., Jang, J., Jung, S., Kim, S., Lim, H., **Yang, H.**, and Yoo, H. *A Study on the Quantification and Improvement of Forest Water Yield for Sustainable Supply of Freshwater.* National Institute of Forest Science: Seoul, South Korea, 2021. (ISBN: 9791160195507)
- Choi, H.T., Kim, J., Lim, H., **Yang, H.**, and Yoo, H. *Development and Utilization of the Forest Water Map.* National Institute of Forest Science: Seoul, South Korea, 2020. (ISBN: 9791160194999)

- Kim, J., Choi, H.T., Lim, H., **Yang, H.**, and Yoo, H. *Analysis on the Current Status of River Survey Methodology for the Effective Investigation of Forest Drought.* National Institute of Forest Science: Seoul, South Korea, 2020. (ISBN: 9791160194913)
- Choi, H.T., Lim, H., and **Yang, H.** *Development Method and Application of Forest Water Map.* National Institute of Forest Science: Seoul, South Korea, 2019. (ISBN: 9791160193800)

CONFERENCE PRESENTATIONS

[Oral Presentations]

- **Yang, H.**, Li, Z., Lee, J.W., Kim, T., Le, P.V.V. "Multi-GPU Training of Geometry-Aware Neural Operators for Cross-Regional Storm Surge Prediction" TACC Symposium for Texas Researchers 26, Texas, USA, Sep. 21-22, 2026.
- **Yang, H.**, Lee, J.W., Santos Cruz, A.U., Subgranon, A., Klepac, S.P., Jiao, J. "Can Hurricane, Surge, and Damage Data Assist in Reconstructing Structural Features?" 10th Young Coastal Scientists and Engineers Conference - Americas, Quebec, Canada, Jun. 4-6, 2024.
- **Yang, H.**, Lee, J.W. "Towards Resilient Texas: Rapid Storm Surge Prediction Using Machine Learning" Planet Texas 2050 Symposium, Texas, USA, Feb. 27-29, 2024.
- **Yang, H.**, Choi, H.T., and Lim, H. "Estimating Flood Peaks in Ungauged Forested Catchments" Joint Symposium on Forest Science, Seoul, Korea, Feb. 15-16, 2023.
- **Yang, H.**, Lim, H., Li, Q., Nam, S., Choi, H.T., Kim, J., and Moon, H. "Developing Flash Flood Predictive Model Using LSTM Neural Networks for Mountainous Areas" Proceeding of the International Symposium on Forest Science, Daegu, Korea, Aug. 24-26, 2022.
- **Yang, H.**, Lim, H., Choi, H.T., and Lee, J. "Development of Throughfall Simulation Models and Prediction Uncertainty Estimation by Different Forest Stand Characteristics" EGU General Assembly 2022, Vienna, Austria, May 23-27, 2022. *Online.*
- **Yang, H.**, Lim, H., Choi, H.T., Kim, J., and Nam, S. "Comparing Machine Learning Algorithms for Developing Optimal PTF and DSM Models in South Korea" Joint Symposium on Forest Science, Seoul, Korea, Feb. 9-10, 2022.
- **Yang, H.**, Lim, H., Choi, H.T., and Kim, J. "Identifying the Characteristics of Forest Water Cycle in Different Forest Stands and Rainfall Events" Proceedings of the International Symposium on Forest Science, Pyeongchang, Korea, Aug. 18-20, 2021. *Online.*
- **Yang, H.**, Lim, H., Choi, H.T., and Kim, J. "Estimating the Spatial Distribution of Forest Soil Water Retention using the Machine Learning Model" Joint Symposium on Forest Science, Seoul, Korea, Feb. 17-18, 2021. *Online.*
- **Yang, H.**, Choi, H.T., and Lim, H. "Pedo-Transfer Functions (PTFs) Development for the Saturated Hydraulic Conductivity of Forest Soil" Proceedings of the International Symposium on Forest Science, Seoul, Korea, Aug. 19-20, 2020. *Online.*
- **Yang, H.**, Choi, H.T., and Lim, H. "Hydraulic Properties of Forest Soils in Different Soil Texture" Proceeding of Summer Meeting of the Korean Society of Forest Science, Seoul, Korea, Aug. 29, 2019.
- **Yang, H.**, Choi, H.T., and Lim, H. "Effects of Forest Thinning on the Water Yield Increase Based on the Long-term Rainfall and Streamflow Monitoring" Joint Symposium of Forest Science, Jinju, Korea, Feb. 20-21, 2019.
- **Yang, H.**, Choi, H.T., and Lim, H. "Selection of the Proper Baseflow Recession Constant Estimation Methods for the Reasonable Hydrograph Analysis in Small Forest Catchments" Joint Symposium of Forest Science, Seoul, Korea, Aug. 29-30, 2018.
- **Yang, H.**, Lee, S., and Im, S. "Development of Estimation Formula for the Mean Stream Velocity in Steep Mountain with Salt-dilution Method" Joint Symposium of Forest Science, Chuncheon, Korea, Feb. 21, 2018.

[Poster Presentations]

- **Yang, H.**, Lee, J.W., Lee, W. "Development of a Physics-Informed Deep Learning Model for Hurricane-Induced Compound Flooding Prediction" HydroML 2026, Texas, USA, May 19-21, 2026.
- **Yang, H.**, Lee, J.W., Lee, W. "Spatiotemporally Heterogeneous Improvements in Compound Flooding Prediction through Physics-Constrained Learning and Their Implications for Coastal Risk" Project SHIELD Workshop, Texas, USA, May 7, 2026.
- Kim, M., **Yang, H.**, Chowdhury, M.E., Lee, J.W. "Compound Flooding and Power Grid Resilience in Puerto Rico: Evaluating Extreme Forcing Duration using Mixed Hurricane Irma and Maria Scenarios" 2026 Energy Research Poster Competition, Texas, USA, Apr. 6-7, 2026.
- **Yang, H.**, Lee, J.W., Santos Cruz, A.U. "A Hybrid AI-Driven and Static Flood Modeling for Real-Time Peak Storm Surge Inundation Mapping" American Geophysical Union (AGU) Fall Meeting 2025, Louisiana, USA, Dec. 15-19, 2025.
- **Yang, H.**, Lee, J.W., Santos Cruz, A.U. "A Hybrid Approach Integrating Machine Learning and Static Flood Models for Efficient Real-Time Peak Storm Surge Inundation Mapping" Artificial Intelligence and Digital Twins for Earth Systems, Texas, USA, Sep. 22-24, 2025.
- **Yang, H.**, Lee, J.W. "Storm Surge Hazard Estimation for Texas Coastal Areas Using Machine Learning" 2024 CAEE Graduate Symposium, Texas, USA, Feb. 23, 2024.

- **Yang, H.**, Moon, H., Lim, H., Kim, J., and Choi, H.T. "Expected Amount of Data Accumulation to Develop Reasonable Flash Flood Predictive Models Using Machine Learning Approaches" American Geophysical Union (AGU) Fall Meeting 2022, Illinois, USA, Dec. 12-16, 2022. *Online.*
- **Yang, H.**, Lim, H., Choi, H.T., Kim, J., and Nam, S. "A Novel Method for Selecting Priority Soil Sample Sites using Prediction Uncertainty of Machine Learning Model Based on Geospatial Information" Frontiers in Hydrology Meeting 2022, San Juan, Puerto Rico, Jun. 19-24, 2022. *Online.*
- **Yang, H.**, Lim, H., Kim, J., and Choi, H.T. "Effects of Soil Topographical Properties on the Forest Soil Field Capacity using Machine Learning Approach" AGU Fall Meeting 2021, Louisiana, USA, Dec. 13-17, 2021. *Online.*
- **Yang, H.** and Im, S. "Salt-dilution Velocity Prediction in Naturalized Channels" SNU and U. of Tokyo Joint Workshop, Seoul, Korea, May 9, 2017.
- **Yang, H.** and Im, S. "Flow Measurement of Mountain Rivers using Salt-dilution Method" The 7th International Symposium of the Asian University Forests Consortium, Hokkaido, Japan, Oct. 11-14, 2016.

PEER-REVIEW SERVICE

- Journal of Hydrology
- Geomatics, Natural Hazards and Risk

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, ArcGIS, QGIS
- **Modeling:** Geometry-aware neural operators, hydrodynamic models (SFINCS, ADCIRC), physics-constrained deep learning, diffusion models
- **High-Performance Computing:** GPU-accelerated model training and inference on HPC systems; multi-GPU and distributed training workflows
- **Experimental and Field:** Installation of hydrological instruments and data collection via datalogger, water level gauge, sap flow sensor, and automatic water sampler; salt-dilution stream discharge measurement; forest soil sampling and hydraulic property analysis

REFERENCES

Jun-Whan Lee, Ph.D.
Assistant Professor
Maseeh Department of Civil, Architectural and Environmental Engineering
University of Texas at Austin
Email: junwhanlee@austin.utexas.edu

Phong V. V. Le, Ph.D.
R&D Staff Member
Environmental Sciences Division
Oak Ridge National Laboratory
Email: lepv@ornl.gov

Wonhyun Lee, Ph.D.
Research Assistant Professor
Bureau of Economic Geology
University of Texas at Austin
Email: wonhyun.lee@beg.utexas.edu